

Morgan Patel

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EDUCATION

Ridgeway University | Ph.D. candidate in Statistics | Metro City, USA | Sep 2021 - Expected May 2026

Ridgeway University | B.S. in Mathematics | Metro City, USA | Sep 2016 - May 2020

EXPERIENCE

Northpeak Capital | Quantitative Research Intern | Metro City, USA | Jun 2025 - Aug 2025

- Built a short-horizon order-book imbalance signal for liquid index futures that raised the desk book's Sharpe ratio from 1.1 to 1.5 over 18 months of out-of-sample backtest after costs.
- Cut the signal's daily turnover from 34% to 21% with a cost-aware position smoother, keeping 90% of gross returns and lowering estimated slippage by a third.
- Validated the signal on 6 years of tick data with purged walk-forward splits and an embargo period, so every backtest decision used only information available at the time.
- Wrote a feature store for 120 microstructure features with point-in-time joins, which the research team reused in two later signal projects.
- Presented the signal, its capacity estimate and failure cases to the portfolio managers, who approved a small live allocation for the next quarter.
- Documented the backtest assumptions, transaction-cost model and known failure regimes in the desk's research wiki, which the next intern cohort used to onboard in their first week.

Ridgeway University | Research Assistant, Statistical Learning Lab | Metro City, USA | Jun 2020 - Aug 2021

- Built the lab's simulation pipeline on the shared cluster, cutting a 2,000-run Monte Carlo study from 3 days to 5 hours, with every run reproducible from its seed and configuration file.
- Derived a variance bound for a sparse regression estimator that tightens the previous bound by a log factor; it is now Section 3 of a paper under review at JASA.
- Taught weekly recitations for 60 students in graduate probability, writing 12 problem sets and earning a 4.8/5 teaching rating.
- Released an open-source R package for high-dimensional covariance estimation with shrinkage and factor models, downloaded 3,000 times in its first year.

PROJECTS

Volatility Forecasting Study | Independent Research | Python, PyTorch | Jan 2024 - Present

- Beat a HAR-RV baseline's out-of-sample QLIKE loss by 7% on 30 equity indices with a temporal convolutional model trained on realized-volatility features.
- Tested significance with Diebold-Mariano tests across the 30 indices; after a Holm correction the forecast gain held at the 5% level on 24 of them and in both crisis periods.
- Wrote up the method, robustness checks and results as a 12-page working paper, which was presented at the department's financial econometrics seminar.

Kaggle Market Prediction Competition | Team of 3 | Python | Mar 2023 - Jun 2023

- Placed 41st of 2,900 teams with a gradient-boosting ensemble on 300 engineered features, finishing in the top 2% of the private leaderboard.
- Cut validation leakage by switching to time-grouped folds, which closed a 0.02 gap between local validation and leaderboard scores.
- Wrote the team's feature-selection script using permutation importance, trimming 900 candidate features to 300 without losing validation score.

SKILLS

Programming: Python, R, PyTorch

Methods: time-series econometrics, high-dimensional statistics, gradient boosting